

Linear constraints

Basic directions

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Optimization: principles and algorithms

Basic directions

The concept of basic and non basic variables allowed us to identify the vertices of the polyhedron

Let's now use them to identify feasible directions

Feasible direction

Reminder

d is feasible at x if

$$\begin{aligned}Ad &= 0, \\ d_i &\geq 0 \text{ if } x_i = 0.\end{aligned}$$

$$x = \begin{pmatrix} x_B \\ x_N \end{pmatrix} = \begin{pmatrix} B^{-1}b \\ 0 \end{pmatrix} \quad d = \begin{pmatrix} d_B \\ d_N \end{pmatrix}$$

$$d_N = (0, 0, 0, 1, 0, 0, 0)$$

$$Ad = Bd_B + Nd_N$$

$$= Bd_B + \sum_{j=m+1}^n A_j d_j = Bd_B + A_p = 0,$$

$$d_B = -B^{-1}A_p.$$

Feasible direction

Theorem 3.44

If x is non degenerate, any basic direction is feasible.

Idea

d is feasible at x if

$$\begin{aligned}Ad &= 0, \\ d_i &\geq 0 \text{ if } x_i = 0.\end{aligned}$$

If x is non degenerate, only non basic variables are 0.

Summary

Using the idea of the basis, the only degrees of freedom are the non basic components of the direction

We set all of them to 0, except one that is set to 1.

We calculate the basic components to verify the equality constraints

If x is non degenerate, it is a feasible direction.

If not, there is no guarantee that it is feasible.